

CURRICULUM VITA

Hacon Christopher

Born: February 14, 1970, Manchester England

Address: Department of Mathematics, University of Utah
155 South 1400 East, JWB 233
Salt Lake City, Utah 84112-0090

Education:

Ph.D.	University of California Los Angeles	Mathematics	1998
M.S.	University of California Los Angeles	Mathematics	1995
B.A.	University of Pisa	Mathematics	1992
Diploma	Scuola Normale Superiore di Pisa	Mathematics	1992

Study and research support:

A.M.S. Centennial Fellowship 2006;
N.S.F. Grant 2005;
A. P. Sloan Fellowship 2003;
N.S.A. Junior Research Grant 2003;
C.N.R. Support for visit to the University of Pisa 2000;
Univ. of Utah Math. Dept. Instructor Award 2000;
C.N.R. Research Fellowship 1998;
UCLA Dissertation Scholarship 1997-1998;
C.N.R. Ph.D. Scholarship 1995-1997;
I.N.d.A.M. Ph.D. Scholarship 1993-1995.

Employment:

2005-present Associate Professor, University of of Utah
2002-2005 Assistant Professor, University of of Utah
2000-2002 Assistant Professor, University of California, Riverside
1998-2000 Mathematics Instructor, University of Utah

Publications:

[1] *Examples of spanned and ample vector bundles with small numerical invariants.* C. R. Acad. Sci. Paris Ser. I Math **323** no. 9, 1025–1029 (1996).

[2] *Remarks on Seshadri constants of vector bundles.* Annales de l'Institut Fourier, Tome 50 (2000) - Fascicule 3.

- [3] *Divisors on principally polarized abelian varieties.* Compositio Mathematica **119** No. 3, December 1999.
- [4] *Fourier transforms, generic vanishing theorems and polarizations of abelian varieties.* Mathematische Zeitschrift **235** (2000) 717-726.
- [5] *Characterization of Abelian Varieties.* (With J. A. Chen) Inventiones Mathematicae **143** (2001) 2, 435-447.
- [6] *Pluricanonical maps of varieties of maximal Albanese dimension.* (With J. A. Chen) Math. Annalen. **320** (2001) 2, 367-380.
- [7] *On algebraic fiber spaces over varieties of maximal Albanese dimension.* (With J. A. Chen) Duke Math. Jour. **111**, issue 1, 159-175
- [8] *Quantum invariant theory and quantum Schubert varieties.* (With R. Fiorese) Jour. of Algebra, **242**, No. 2, Aug 2001, 433-446.
- [9] *Deformations of flat line bundles and their metrics.* (With H. Clemens) Amer. Journal of Math. 124.4 August 2002, 769-815
- [10] *Surfaces with $P_g = q = 3$.* (With R. Pardini) Trans. of the Amer. Math. Soc. **354** (2002), 2631-2638.
- [11] *On the birational geometry of varieties of maximal Albanese dimension.* (With R. Pardini) Journ. Reine Angew. Math. **546** (2002)
- [12] *Effective criteria for birational morphisms.* Jour. London Math. Soc. Vol. 67 part 2 April 2003, 337-348
- [13] *On the irregularity of the image of the Iitaka fibration.* (With J. A. Chen) To appear in Comm. in Algebra Vol. 32, No. 1, pp. 203-215 (2004)
- [14] *Varieties with $P_3 = 3$ and $q(X) = \dim(X)$.* Math. Nachr. **278**, No. 4, 409-420 (2005).
- [15] *Linear series of irregular varieties.* (With J. A. Chen) Proceedings of the symposium on Algebraic Geometry in East Asia. World Scientific (2002)
- [16] *On infinite dimensional Grassmannians and their quantum deformations.* (With R. Fiorese) Rendiconti del Seminario Matematico dell'Università di Padova Vol. 111 (2004)
- [17] *A derived category approach to generic vanishing theorems.* Journ. Reine Angew. Math. **575** (2004)
- [18] *Varieties with $P_3 = 4$ and $q(X) = \dim(X)$.* (With J. A. Chen) Annali della Scuola Normale Superiore di Pisa, Classe di Scienze, serie V, Vol. III, 2 (2004).
- [19] *An example of surface of general type with $p_g = q = 2$ and $K_X^2 = 5$.* (With J. A. Chen) Pac. Jour. Math. Vol 233, No. 2, (2006)
- [20] *Birational characterization of products of curves of genus 2.* (With R. Pardini) Math. Research Letters **12**, no. 1, (2005)
- [21] *On the degree of the canonical maps of 3-folds.* Proc. Japan Acad. Ser. A Math. Sci. **80** (2004), no. 8, 166-167.

- [22] *Holomorphic one forms on varieties of general type*. (Joint with S. Kovács) Ann. Sci. Ecole Norm. Sup. (4) 38 (2005), no. 4, 599–607
- [23] *Singularities of divisors of low degree on abelian varieties*. (Joint with O. Debarre) To appear in Math. Zeit.
- [24] *Boundedness of pluricanonical maps of varieties of general type*. (Joint with J. McKernan) Invent. Math. 166, 1-25 (2006)
- [25] *Shokurov's rational connectedness conjecture*. (Joint with J. McKernan) To appear in Duke Math. J. math.AG/0504330
- [26] *On the existence of flips*. (Joint with J. McKernan) math.AG/0507597
- [27] *On the geography of Gorenstein minimal 3-folds of general type* (Joint with M. Chen) To appear in the Asian Journal of Mathematics.
- [28] *Extension theorems and the existence of flips*. (Joint with J. McKernan) To appear in flips for 3-folds and 4-folds, A. Corti editor.
- [29] *Pluricanonical systems on irregular 3-folds of general type*. (With J.A. Chen) To appear in Math. Zeit.
- [30] *Termination of (many) 4-dimensional log flips*. (With V. Alexeev and Y. Kawamata) math.AG/0605137
- [31] *Existence of minimal models for varieties of general type* (Joint with Birkar, Cascini, McKernan) math.AG/0610203

Invited Lectures:

Algebraic Geometry Seminars: Univ. of Mich. Ann Arbor 1997 and 2004, University of Illinois Chicago 1997, UC Irvine 1997, Princeton 1999 and 2005, Univ. of Utah 2001 and 2002, UC Santa Barbara 2001, Harvard-MIT 2002 and 2006, Univ. of Washington 2003, Univ. Urbana Champaign 2004, Stanford 2004, SUNY Stony Brook 2005

Colloquiums: Univ. of Utah 2001, UC Riverside 2001, Princeton 2005, SUNY Stony Brook 2005

Conferences: Western Algebraic Geometry Seminar, Riverside 1997, AMS meeting in Notre Dame, Indiana 2000, Southern California Algebraic Geometry Workshop, UC Santa Barbara, 2000, AMS meeting in Salt Lake City 2002, 2004 NCTS Workshop in Algebraic Geometry (Taiwan), CIRM summer school (5 lectures) in Trento Italy, June 2003, WAGS meeting, Univ. of Washington 2005, 2005 AMS Summer Institute on Algebraic Geometry in Seattle, 2005 Algebraic Geometry Workshop, Univ. Michigan, 2006 Recent trends in Algebraic Geometry, Banff

Other professional activities and service:

Refereed for: i) The proceedings of the conference in celebration of Herb Clemens' 60th birthday, to be published by the AMS. ii) Journ. Reine Angew. Math. iii) Nagoya Math. Jour. iv) Mathematical Research Letters v) Journal of Algebra vi) Topology vii) Trans. American Math. Soc. viii) Adv. in Math. ix) Snowbird conference proceedings x) Annals of Math. xi) Inventiones xii) Jour. Amer. Math. Soc. xiii) Comp. Math, xiv) Amer. Jour. Math, xv) Proc American Math. Soc., xvi) Acta Math.;

Conference/courses organized: i) Recent Trends in Algebraic Geometry: session of the 981st AMS meeting, Slat Lake City, 2002. ii) Arc spaces and motivic integration: VIGRE summer mini-course at the Univ. of Utah. iii) WAGS Winter 2005, iv) WAGS Winter 2006.

Committees and other service: Undergraduate advisor (UCR 2000-2001); Represented the Univ. of Utah Math Dept. at the statewide advisor meeting, winter 02-03; Committee for 5 year evaluation of Mladen Bestvina, Spring 02-03; Reader for Junior Science and Humanities symposium 03, 04, 05, 06; Annuities and Salary Committee 04-05; Graduate committee 04-05; Undergraduate Awards and Scholarships Committee 03-05; Pi-Mu-Epsilon faculty correspondent 04-05; Ad Hoc RPT Committee for Y.P. Lee 05; Orals: A. Bean, G. Todorov, Q. Song, C. Elfriz, S. Kitchen; Defenses: F. Sato.

Courses Taught:

1998-1999:	Math 2250	ODEs and Linear Algebra
	Math 2210	Calculus III
	Math 1050	College Algebra
1999-2000:	Math 2250	ODEs and Linear Algebra
	Math 1210	Calculus I
	Math 7800	Topics in Algebraic Geometry
2000-2001:	Math 010A	Calculus: Several Variables
	Math 113	Applied Linear Algebra
	Math 171	Modern Algebra: Introduction
	Math 009B	First-Year Calculus
2001-2002:	Math 171	Modern Algebra: Introduction
	Math 172	Modern Algebra
	Math 010A	Calculus: Several Variables
	Math 009A	First-Year Calculus
2002-2003:	Math 1030	Quantitative Thinking
	Math 4030	Foundations of Algebra
	Math 5520	Intr. Algebra/Geom. Topology

2003-2004:	Math 7800	Topics in Algebraic Geometry
	Math 1030	Quantitative Thinking
2004-2005:	Math 6310	Algebra
	Math 1030	Quantitative Thinking
	Math 6320	Algebra
2005-2006:	Math 7800	Topics in Algebraic Geometry
	Math 1030	Quantitative Thinking
	Summer High School Program	